

- copulation, the vagina accommodates the male's penis and receives sperm.
- valence** The bonding capacity of a given atom; the number of covalent bonds that an atom can form, which usually equals the number of unpaired electrons in its outermost (valence) shell.
- valence electron** An electron in the outermost electron shell.
- valence shell** The outermost energy shell of an atom, containing the valence electrons involved in the chemical reactions of that atom.
- van der Waals interactions** Weak attractions between molecules or parts of molecules that result from transient local partial charges.
- variable** A factor that varies in an experiment.
- variation** Differences between members of the same species.
- vas deferens** In mammals, the tube in the male reproductive system in which sperm travel from the epididymis to the urethra.
- vasa recta** The capillary system in the kidney that serves the loop of Henle.
- vascular cambium** A cylinder of meristematic tissue in woody plants that adds layers of secondary vascular tissue called secondary xylem (wood) and secondary phloem.
- vascular plant** A plant with vascular tissue. Vascular plants include all living plant species except liverworts, mosses, and hornworts.
- vascular tissue** Plant tissue consisting of cells joined into tubes that transport water and nutrients throughout the plant body.
- vasectomy** The cutting and sealing of each vas deferens to prevent sperm from entering the urethra.
- vasoconstriction** A decrease in the diameter of blood vessels caused by contraction of smooth muscles in the vessel walls.
- vasodilation** An increase in the diameter of blood vessels caused by relaxation of smooth muscles in the vessel walls.
- vasopressin** See antidiuretic hormone (ADH).
- vector** An organism that transmits pathogens from one host to another.
- vegetal pole** The point at the end of an egg in the hemisphere where most yolk is concentrated; opposite of animal pole.
- vegetative propagation** Asexual reproduction in plants that is facilitated or induced by humans.
- vegetative reproduction** Asexual reproduction in plants.
- vein** (1) In animals, a vessel that carries blood toward the heart. (2) In plants, a vascular bundle in a leaf.
- ventilation** The flow of air or water over a respiratory surface.
- ventral** In an animal with bilateral symmetry, pertaining to the underside (in most animals) or front (in animals with upright posture) of the body.
- ventricle** (ven'-tri-kul) (1) A heart chamber that pumps blood out of the heart. (2) A space in the vertebrate brain, filled with cerebrospinal fluid.
- venule** (ven'-yul) A vessel that conveys blood between a capillary bed and a vein.
- vernalization** The use of cold treatment to induce a plant to flower.
- vertebrate** A chordate animal with vertebrae, the series of bones that make up the backbone.
- vesicle** (ves'-i-kul) A membrane-bound sac in or outside a cell.
- vessel** A continuous water-conducting micro-pipe found in most angiosperms and a few nonflowering vascular plants.
- vessel element** A short, wide, water-conducting cell found in the xylem of most angiosperms and a few nonflowering vascular plants. Dead at maturity, vessel elements are aligned end to end to form micropipes called vessels.
- vestigial structure** A feature of an organism that is a historical remnant of a structure that served a function in the organism's ancestors.
- villus** (plural, villi) (1) A finger-like projection of the inner surface of the small intestine. (2) A finger-like projection of the chorion of the mammalian placenta. Large numbers of villi increase the surface areas of these organs.
- viral envelope** A membrane, derived from membranes of the host cell, that cloaks the capsid, which in turn encloses a viral genome.
- virulent phage** A phage that replicates only by a lytic cycle.
- virus** An infectious particle incapable of replicating outside of a cell, consisting of an RNA or DNA genome surrounded by a protein coat (capsid) and, for some viruses, a membranous envelope.
- visceral mass** One of the three main parts of a mollusc; the part containing most of the internal organs. See also foot and mantle.
- visible light** That portion of the electromagnetic spectrum that can be detected as various colors by the human eye, ranging in wavelength from about 380 nm to about 740 nm.
- vital capacity** The maximum volume of air that a mammal can inhale and exhale with each breath.
- vitamin** An organic molecule required in the diet in very small amounts. Many vitamins serve as coenzymes or parts of coenzymes.
- viviparous** (vi-vip'-uh-rus) Referring to a type of development in which the young are born alive after having been nourished in the uterus by blood from the placenta.
- voltage-gated ion channel** A specialized ion channel that opens or closes in response to changes in membrane potential.
- vulva** Collective term for the female external genitalia.
- water potential** (ψ) The physical property predicting the direction in which water will flow, governed by solute concentration and applied pressure.
- water vascular system** A network of hydraulic canals unique to echinoderms that branches into extensions called tube feet, which function in locomotion and feeding.
- wavelength** The distance between crests of waves, such as those of the electromagnetic spectrum.
- wetland** A habitat that is inundated by water at least some of the time and that supports plants adapted to water-saturated soil.
- white matter** Tracts of axons within the CNS.
- whole-genome shotgun approach** Procedure for genome sequencing in which the genome is randomly cut into many overlapping short segments that are sequenced; computer software then assembles the complete sequence.
- wild type** The phenotype most commonly observed in natural populations; also refers to the individual with that phenotype.
- wilting** The drooping of leaves and stems as a result of plant cells becoming flaccid.
- wobble** Flexibility in the base-pairing rules in which the nucleotide at the 5' end of a tRNA anticodon can form hydrogen bonds with more than one kind of base in the third position (3' end) of a codon.
- xerophyte** (zir'-o-fit') A plant adapted to an arid climate.
- X-linked gene** A gene located on the X chromosome; such genes show a distinctive pattern of inheritance.
- X-ray crystallography** A technique used to study the three-dimensional structure of molecules. It depends on the diffraction of an X-ray beam by the individual atoms of a crystallized molecule.
- xylem** (zi'-lum) Vascular plant tissue consisting mainly of tubular dead cells that conduct most of the water and minerals upward from the roots to the rest of the plant.
- xylem sap** (zi'-lum) The dilute solution of water and minerals carried through vessels and tracheids.
- yeast** Single-celled fungus. Yeasts reproduce asexually by binary fission or by the pinching of small buds off a parent cell. Many fungal species can grow both as yeasts and as a network of filaments; relatively few species grow only as yeasts.
- yolk** Nutrients stored in an egg.
- zero population growth (ZPG)** A period of stability in population size, when additions to the population through births and immigration are balanced by subtractions through deaths and emigration.
- zona pellucida** The extracellular matrix surrounding a mammalian egg.
- zoned reserve** An extensive region that includes areas relatively undisturbed by humans surrounded by areas that have been changed by human activity and are used for economic gain.
- zone of polarizing activity (ZPA)** A block of mesoderm located just under the ectoderm where the posterior side of a limb bud is attached to the body; required for proper pattern formation along the anterior-posterior axis of the limb.
- zoonotic pathogen** A disease-causing agent that is transmitted to humans from other animals.
- zoopagomycete** A member of the fungal phylum Zoopagomycota, multicellular parasites or commensal symbionts of animals; sexual reproduction, where known, involves the formation of a sturdy structure called a zygosporangium.
- zoospore** Flagellated spore found in chytrid fungi and some protists.
- zygomycete** (zi'-guh-mi'-set) A member of the fungal phylum Zygomycota, characterized by the formation of a sturdy structure called a zygosporangium during sexual reproduction.
- zygosporangium** (zi'-guh-spör-an'-jē-um) (plural, **zygosporangia**) In zygomycete fungi, a sturdy multinucleate structure in which karyogamy and meiosis occur.
- zygote** (zi'-göt) The diploid cell produced by the union of haploid gametes during fertilization; a fertilized egg.